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Professional Summary

Senior Software Engineer with 16+ years of experience designing and delivering scalable enterprise and cloud-native systems across finance, insurance, healthcare, and SaaS industries. Currently completing a Master of Software Engineering in New Zealand, specialising in software architecture, AI systems, and cloud engineering.

Experienced in building API-first platforms using .NET and Python (FastAPI), applying Clean Architecture, SOLID principles, and DevOps best practices. Proven track record of modernising legacy systems, improving performance, and delivering production-ready solutions that reduce operational overhead and increase reliability. Passionate about building ethical, high-impact technology and mentoring high-performing teams.

Key Skills

AI & Data: Python, FastAPI, LLM Integration, FAISS Vector Search, OCR, Data Analysis, Exploratory Analysis

Languages & Frameworks: C#, VB.Net, .NET Core, .NET Framework, TypeScript, JavaScript, Angular, Bootstrap, Kendo UI

Web & API Development: Android (Kotlin), Angular, APIs, ASP.NET MVC, REST, SOAP, WCF

Cloud & DevOps: AWS Lambda, API Gateway, RDS, Oracle Cloud, Docker, CI/CD, GitHub Actions, Microsoft Azure, Kubernetes, Azure DevOps, Bamboo

Databases: SQL Server, MySQL, Oracle, SQLite, DB2, SSIS, SSRS, Crystal Reports

Testing & Automation: Xunit, NUnit, SpecFlow, Moq, Selenium, Cypress, MS Unit Test

Design Patterns & Principles: Clean Architecture, Domain-Driven Design, DTO-Mapper-Repository, SOLID, Microservices, OOP, Repository & Unit of Work Patterns, MVC, MVVM, SOA

Methodologies: Agile (SCRUM), Kanban, Waterfall

Tools: Git, TFS, SVN, Jira, Confluence, Visual Studio, PDF Pro Reports, Photoshop

Application Analyst Developer

Satellite Office – Smartgroup Australia | *Jul 2023 – Jul 2025* Smartgroup provides enterprise-grade salary packaging and fleet management solutions across the corporate, healthcare, government, and education sectors.

- **Developed new applications and maintained legacy systems**, ensuring reliability, compatibility, and ongoing performance improvements across production environments.
- **Optimised and maintained stored procedures**, reducing data-processing time from **23 hours to 2–3 hours** through indexing, query tuning, and batch refactoring.
- **Developed and enhanced SSRS and PDF Pro reports**, delivering critical data insights to various business units and improving operational visibility.

- **Led feature development and enhancements** to existing systems, introducing new business functionalities and improving overall user experience.
- **Troubleshoot and investigated application and data-related issues**, resolving incidents efficiently to minimise downtime and ensure system stability.
- **Provided technical support** to internal and external clients, addressing tickets, user issues, and change requests effectively.
- **Participated in the code review process**, implementing and enforcing coding standards and **best practices** across multiple development teams.
- **Created and maintained unit tests, integration tests, and automated test suites** for both new and legacy applications, ensuring software reliability and regression safety.
- **Collaborated closely with project managers, business analysts, and end users**, translating business requirements into high-quality technical solutions.
- **Researched new tools, frameworks, and technologies**, integrating modern approaches to enhance project efficiency and maintain industry best practices.

Senior Software Engineer

Willis Towers Watson (Philippines) | *Mar 2020 – Jul 2023* Developed and supported enterprise-scale HR and benefits platforms including **BrightChoices**, **Benefit Access**, and **FSA Portal**, enabling employees worldwide to manage customised insurance and wellness packages.

- **Led feature development and enhancement work** for complex enterprise applications using **.NET Core, Angular, and SQL Server**, ensuring quality, scalability, and maintainability.
- **Participated in the extraction and conversion of monolithic components into microservices**, improving system modularity, deployment speed, and performance.
- Designed and implemented new APIs, data modules, and backend services that integrated seamlessly with existing platforms.
- Applied **SOLID principles, Repository, Service Layer, Factory, and DTO** patterns to build clean, maintainable, and scalable systems.
- **Actively participated in code reviews**, providing constructive feedback, maintaining coding standards, and promoting consistency across teams.
- **Mentored and coached junior developers**, strengthening their understanding of clean code, debugging, version control, and Agile delivery.
- **Participated in the technical hiring process**, assessing candidate skills, conducting interviews, and helping select high-performing developers.
- Delivered well-tested, production-ready solutions with **unit, integration, and UI automation tests** using **NUnit, SpecFlow, Selenium, and Cypress**, achieving 98% coverage.
- Championed **best practices**, including dependency injection, CI/CD pipelines, modular architecture, and continuous improvement methodologies.
- Hosted **“Lunch & Learn” sessions** to share emerging frameworks, new tools, and architecture insights across global teams.
- Collaborated with cross-functional Agile squads to deliver secure, performant, and user-focused software aligned with enterprise standards.

.NET Core Developer

AXA Philippines | *Aug 2018 – Mar 2020* AXA is one of the world's leading insurers, providing protection and financial solutions to more than 57 million people globally.

At AXA, I worked on **ACE-DNN – Solutions Delivery**, a generic platform designed to automate backend insurance operations. The goal was to help internal teams move away from long manual processes and create faster, more accurate reporting systems to support client management and decision-making.

- **Developed and integrated new .NET Core modules**, improving overall system performance and user experience.
- **Refactored legacy SQL code and optimised database queries**, reducing average processing time by around **45%**.
- One of my most significant achievements was **automating the manual reporting and data-generation process**. This task used to take the operations team **2–3 months**, but after automation, it was completed in **just one day**, saving time and improving productivity across departments.
- **Participated in Change Advisory Board (CAB) meetings**, presenting planned changes, discussing risks and impacts, and coordinating deployment approvals with senior operations and management teams.
- **Handled on-premises deployments**, ensuring environments were properly configured and that each rollout was successful and issue-free.
- **Conducted code reviews and maintenance activities**, keeping the application stable, secure, and aligned with best practices.
- **Collaborated with QA engineers and business analysts** to identify and resolve issues early in the cycle, improving release quality and reducing rework.
- **Maintained a complete suite of unit and integration tests** with **100% coverage**, which helped the team release confidently with all tests consistently passing.
- **Reviewed and optimised SQL scripts and stored procedures**, enhancing data accuracy and report reliability across different modules.

Full Stack Developer

L.R.B. Medical & Maternity Clinic | *Jan 2018 – Aug 2018* Developed and maintained a **clinic management system** supporting patient records, billing, and scheduling. The project was built using **.NET Core, Angular, SQL Server, and Entity Framework**, improving operational efficiency and reporting accuracy for clinical and administrative staff.

- **Engineered an integrated clinic management platform** using **C#, .NET Core, Angular, and SQL Server 2014**, automating patient records, billing, and reporting workflows.
- **Optimized data flow and PSA birth-record submissions**, cutting manual reporting time by **over 50%** and improving accuracy and auditability.
- **Implemented secure patient-data storage and access layers**, giving doctors real-time health insights for safer, data-driven decision-making.
- **Managed the entire SDLC**, from requirements gathering and architecture design to testing, deployment, and staff onboarding.
- **Improved system performance and usability**, enhancing the clinic's operational reliability and long-term scalability.

C#.NET Developer

Eclaro International Inc. – OneBeacon Insurance Group (USA) | May 2015 – Dec 2017 (OneBeacon Insurance Group is a U.S.-based property and casualty insurance company providing specialized commercial insurance solutions.)

- **Modernized legacy ASP.NET systems** by upgrading front-end components from **HTML4 to HTML5** and integrating **TypeScript** and **Angular**, resulting in a more responsive and maintainable enterprise web platform.
- **Developed and integrated RESTful APIs**, improving data consistency and communication between front-end and back-end systems.
- **Led database migration from Mainframe/DB2 to SQL Server**, optimizing stored procedures, SQL scripts, and indexes for improved performance and reliability.
- **Implemented automated testing frameworks** using **xUnit** and **NUnit**, and **led Selenium test development**, improving overall test coverage, reliability, and regression efficiency.
- **Managed CI/CD pipelines** using **Atlassian Bamboo**, automating build, test, and deployment stages for consistent and faster releases.
- **Enhanced and maintained 100+ SSRS reports**, adding new data insights, improving report accuracy, and creating additional dashboards for executive use.
- **Established and enforced .NET coding standards**, promoting maintainability, scalability, and team-wide consistency.
- **Re-engineered core business modules and reporting pipelines**, achieving up to **35% faster system performance** and greater operational stability.

Lead Software Analyst

Equitable Computer Services Inc. | Mar 2008 – May 2015 | Makati, Philippines (Equitable Computer Services is a technology solutions provider supporting major Philippine financial institutions including Equitable Bank, Robinsons Bank, and BDO.)

- **Led a team of three junior developers**, mentoring them on coding standards, debugging, and full system development processes from requirements gathering to deployment.
- **Designed and developed enterprise-scale banking applications** for **Equitable Bank**, **Robinsons Bank**, and **BDO**, managing both **front-end and back-end** development.
- **Developed the Credit Card Management System from the ground up**, including **database architecture**, **workflow automation**, and **reporting modules** for credit and debit card operations.
- **Owned and maintained backend data processes**, generating critical financial and compliance reports to support credit, debit, and reconciliation systems.
- **Collaborated directly with senior executives (VP-level)** to design and implement **client-facing systems** such as **customer reward and raffle platforms** for banks and insurance partners, enhancing **customer engagement** and **loyalty programs**.
- **Built applications for sister companies** including **myHealth** and **Medilink**, integrating healthcare management and billing systems with financial platforms to streamline patient and partner transactions.
- **Delivered high-volume ASP.NET and Windows applications**, ensuring data security, scalability, and compliance with banking and privacy standards.

Education

Master of Software Engineering (2025 – 2026) *Yoobee College of Creative Innovation – Auckland, New Zealand*

Bachelor of Science in Computer Science (2003 – 2007) *Fillamer Christian University – Roxas City, Philippines*

Projects Handled

Selected professional and academic projects that demonstrate full-stack development, architecture, and problem-solving expertise:

Selected Academic & Personal Projects

KaiHelper – AI-Powered Grocery Budgeting App *Master of Software Engineering – Team Project*

- Designed and built a **production-ready budgeting platform** using **Python FastAPI**, Clean Architecture, DTO-Mapper-Repository pattern, and SOLID principles.
- Implemented **receipt image parsing (OCR + GPT-4o)** to automatically extract grocery items and expenses.
- Built **budget tracking, categories, funding, and insights** modules with clear domain separation.
- Planned deployment using **AWS (Lambda, API Gateway, RDS)** with cost-conscious architecture.
- Integrated **Android (Kotlin) client** consuming REST APIs.
- Emphasised real-world product thinking: scalability, maintainability, and user-centric design.

RecallAI – Personal Knowledge & LLM Assistant *Independent Project*

- Architected a **self-hosted AI assistant** using **FastAPI, local LLMs** (DeepSeek, Qwen, TinyLlama), and **FAISS vector search**.
- Implemented document ingestion, embeddings, semantic search, chat, and multimodal pipelines.
- Deployed on **Oracle Cloud Free Tier VM**, optimising for low-cost, offline-first inference (no paid APIs).
- Designed modular AI services for future expansion (chat, code generation, research, transcription).

Data Analytics, ITS & Quantum Computing (Academic)

- Performed **exploratory and confirmatory data analysis** on public transport datasets (delays, traffic, weather, passenger load).
- Designed **Intelligent Transportation System (ITS)** recommendations using real-world constraints.
- Conducted **quantum computing experiments** (Bell states, Grover's algorithm) using Python-based tooling and simulators.
- Produced research-grade reports aligned with postgraduate assessment standards.

Industrial projects (Work Experience)

- **Salary Packaging & Financial Systems** – Built and maintained enterprise-grade financial platforms at Smartgroup (via Satellite Office), improving scalability, automation, and compliance.
- **BrightChoices / Benefit Access / FSA Portal** – Delivered new features and performance improvements for employee benefits and spending platforms at Willis Towers Watson.
- **ACE Solutions Delivery** – Automated insurance backend operations at AXA Philippines, reducing manual reporting from 2–3 months to 1 day.

- **HR & Healthcare Systems** – Designed and developed HRIS, billing, and healthcare management applications supporting enterprise analytics and business operations.

For a complete list of projects, technical case studies, and live demos, please visit my portfolio:

🌐 retiangson.github.io/portfolio

Professional Interests

- **Software Architecture & System Design** – Passionate about creating scalable, maintainable, and high-performing systems through modern design principles and clean architecture.
- **Cloud Engineering & Microservices** – Interested in building cloud-native, distributed applications using containerisation, DevOps, and continuous integration practices.
- **AI, Quantum Computing & Blockchain Technologies** – Enthusiastic about exploring next-generation technologies that enhance automation, computation, and secure data exchange.
- **Financial & Medical Software Systems** – Deeply interested in developing secure, efficient, and innovative solutions for financial and healthcare industries where precision and reliability matter most.
- **Cultural Learning & Leadership Growth** – Open to learning from New Zealand's professional culture and practices, with a goal to grow into leadership and mentoring roles through local experience.
- **Faith-Inspired Innovation** – Driven to build technology that uplifts people, encourages ethical digital transformation, and contributes positively to communities.
- **Sustainable & Ethical Technology Development** – Committed to creating solutions that balance innovation with environmental and social responsibility.
- **Continuous Learning & Emerging Technologies** – Dedicated to lifelong learning and staying current with evolving tools, frameworks, and methodologies in the software engineering field.

References

Available upon request.